

David Khachatryan - Week 3 Report

Summary

When generating figure 1, we showed the proportional (P-only) controller responding to step-changes in joystick thrust and desired pitch. The motor pairs correctly diverge in opposite directions, showing that our linear-interpolation mapping from joystick to physical thrust and pitch is calibrated correctly..

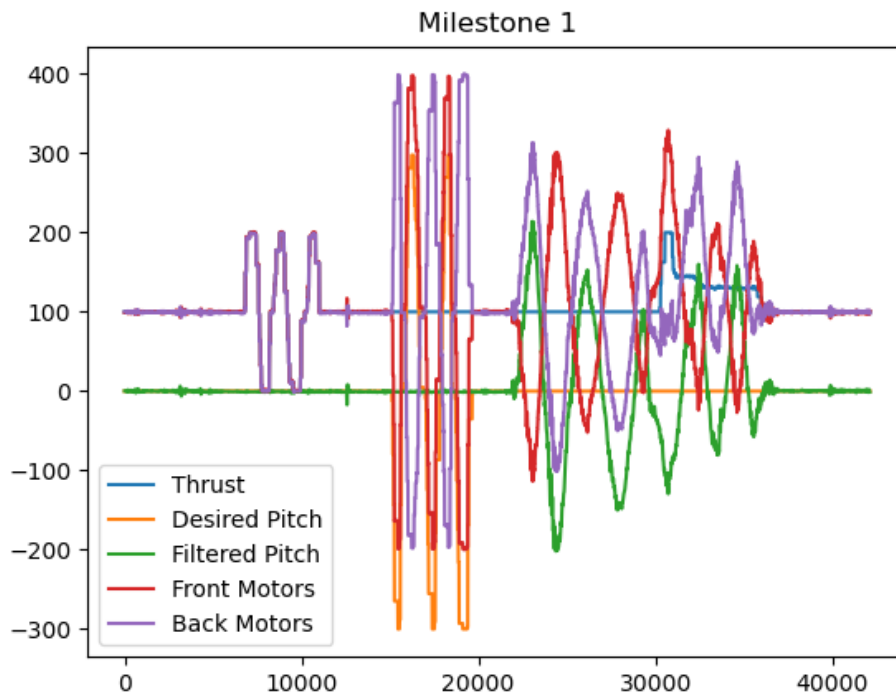


Figure 1: Milestone 1 Graph

Figure 2 adds the derivative term. The graph confirms that measured pitch velocity (gyro Y) is fed back with the correct sign: when the IMU rotates nose-up the back motors are commanded higher, damping the motion.

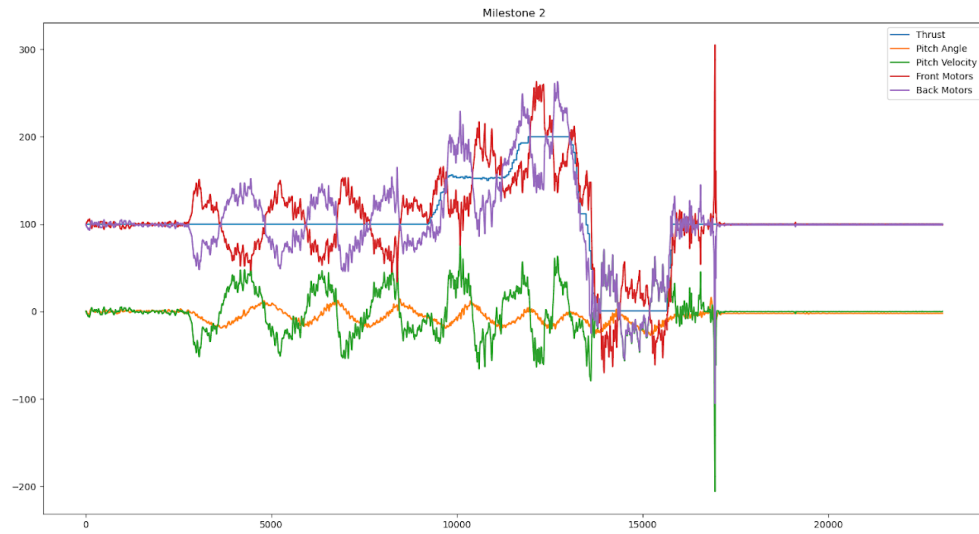


Figure 2: The uptick at the end is due to an accidental drop of the IMU

Figure 3 introduces the integral term. The integrator brings the steady-state error to within ± 1 deg after ≈ 1.5 s. When changing the pitch, front and back motors cross the thrust at the same point.

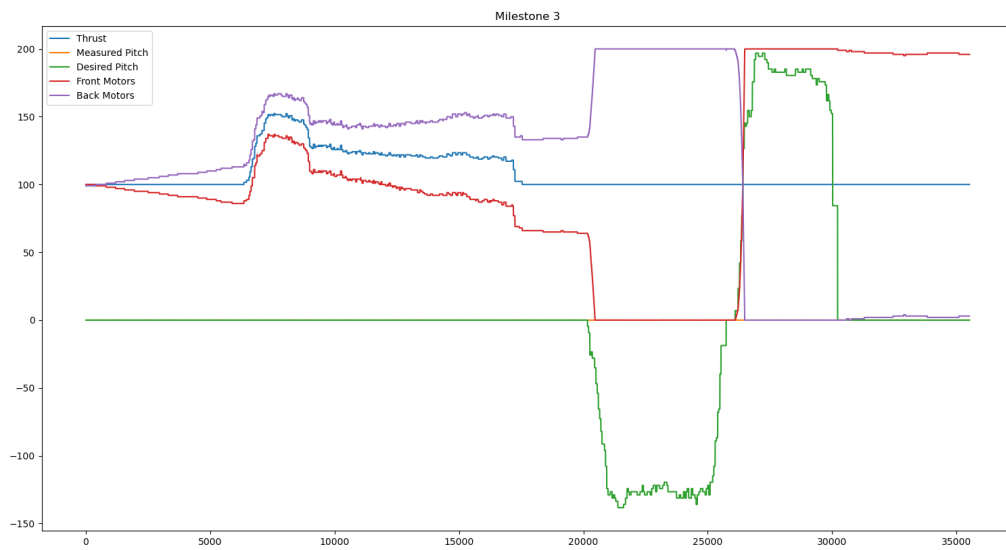


Figure 3: Milestone 3

In Figure 4, we combined all the P, I, and D functions into one formula.

```
motor_commands[0] = thrust + (P_GAIN * p_error) - (D_GAIN * imu_data[5]) + integral_pitch;  
motor_commands[1] = thrust + (P_GAIN * p_error) - (D_GAIN * imu_data[5]) + integral_pitch;  
motor_commands[2] = thrust - (P_GAIN * p_error) + (D_GAIN * imu_data[5]) - integral_pitch;  
motor_commands[3] = thrust - (P_GAIN * p_error) + (D_GAIN * imu_data[5]) - integral_pitch;
```

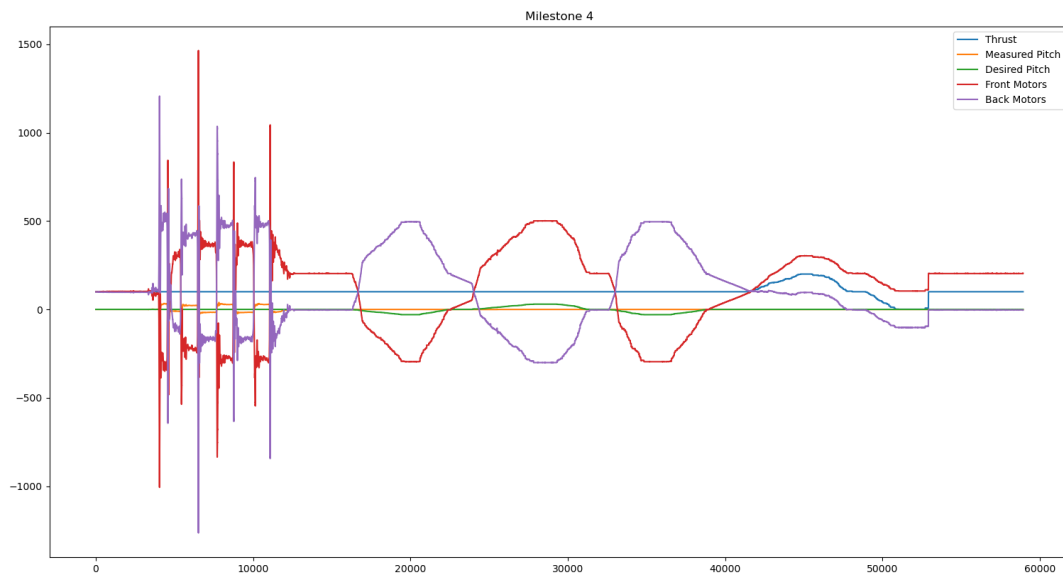


Figure 4: This graph shows it settling at 100 and -100. Drew saw a better settling graph, but we did not save it for submission.

What went well

We worked together to come up with the linear interpolation that correctly converted joystick commands into thrust and pitch. This went pretty smoothly and after about an hour we were trying to graph, thinking that we will finish everything on time.

What did not go well, why?

We spent a lot of time debugging a bouncing issue in our graph. Our motor commands would never go negative, which was really odd, considering that everything else looked correct. After feeling like we were in a limbo, we went over the graph plotting algorithm and it turns out our parsing was not picking up negative commands if the label had more than one word. This went horribly wrong for us and we could not finish the plots during the class.

What will you change for next class

In the next classes, I should pay more attention to the graphing code. It is really frustrating to spend most of the class looking for a bug at the wrong place. Writing tests is one way to avoid this, so going forward more tests would help avoid this sort of problem.

Contributions

I think we contributed to the code equally thanks to the VSCode liveshare extension. The graphing part is entirely my code, so I feel bad that it did not function properly. So I would say I contributed 40 in total and Burke contributed 60.